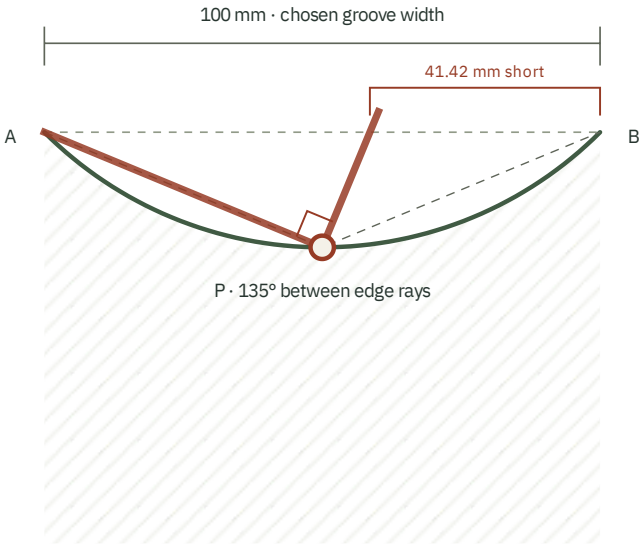
# The cut inside a column

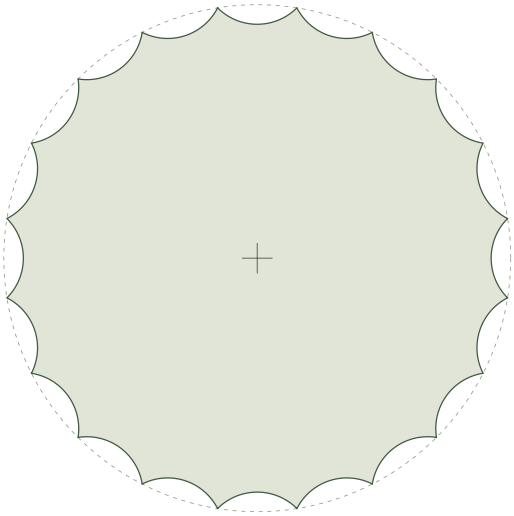
Doric · square and compass. Chosen groove width 100 mm. Tool at 50% of the arc. Ionic fillets 20% of each pitch (a modern choice). All drawings preserve equal horizontal and vertical scales.

## The hollow and the tool




Thin rays reach both edges. The vermilion tool has a fixed 90° corner; its second contact may fall short of B. The gap is measured along A–B.

## The repeated drum section



Real circular hollows and outer-circle fillets. The dashed circle is the uncut shaft. This geometry does not predict structural strength or visual perception.

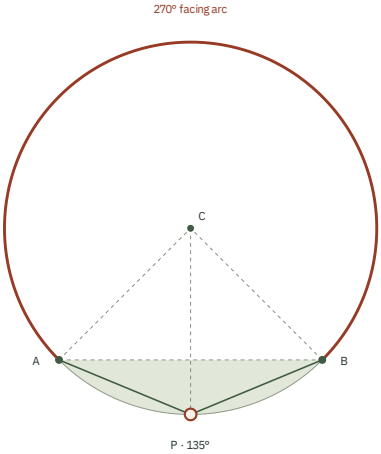
Doric square-and-compass construction: depth = width × (√2 − 1) / 2; edge angle 135°. Ionic semicircle: depth = width / 2; edge angle 90°. The Doric flat-faced alternative has no hollow. Prescriptions do not establish that all surviving ancient profiles follow these curves.

Save the exact coordinates and all three comparisons (CSV). Save the millimetre template (SVG). Print at actual size and check the 25 mm bar; physical printer scaling has not been tested.

# Same chord. Which segment?

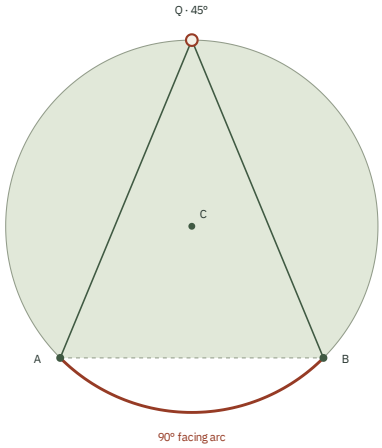
The green region contains the selected point. The orange arc faces it. A quarter-circle hollow puts P in the smaller segment, making 135°, while Q in the larger segment makes 45°. These are modern degree calculations of the chosen model, not ancient measured flute profiles.

From the hollow's arc: P




**135° at P.** Green region: the smaller segment, containing P. Orange arc: the other side of A–B.

From the other arc: Q




**45° at Q.** Green region: the larger segment, containing Q. The chord and circle are unchanged.

Two isosceles triangles explain P.

CA, CP and CB are radii of one circle. In triangles CAP and CPB, the two base angles in each triangle are equal. Their angles at C divide the cut's 90° arc.

(180° – 45.00°) ÷ 2 = **67.50°**

base angle in CAP

(180° – 45.00°) ÷ 2 = **67.50°**

base angle in CPB

At P, add them: **67.50° + 67.50° = 135°**. Moving P changes the two parts but not their sum: 180° – 90° ÷ 2.

A, P, B and Q form a cyclic quadrilateral. Its opposite angles add to two right angles (Euclid III.22), so Q makes **180° – 135° = 45°**.

The arc extents and degree arithmetic are modern annotations. A 270° label describes an arc; it does not assign Euclid an ordinary 270° central angle. The Q comparison is geometry, not another carved-flute prescription. Connecting these texts is our teaching choice; it does not document a mason's method.

Save the exact P/Q comparison on both arcs (CSV).

## Read the region and the comparison.

Ἐν κύκλῳ ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστιν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττονι τμήματι μείζων ὀρθῆς

In circulo angulus in semicirculo positus rectus est, qui autem in segmento maiore positus est, minor recto, qui autem in segmento minore positus est, maior recto

“In a circle, the angle in the semicircle is right; that in the larger segment is less than a right angle, and that in the smaller segment is greater than a right angle.”

Euclid III.31, opening comparisons. Greek edited by J. L. Heiberg; Latin translated by Heiberg (1883), not an ancient Latin version. The English and our degree calculations are new. Inspect the Greek page · Inspect the Latin page.

### ἐν τῷ μείζονι τμήματι

“In the larger segment.” The dative follows *ἐν*. *τμῆμα → τμήματι* names the circular region bounded by a chord and an arc. It is not just the straight line A–B.

### ἐλάττων ὀρθῆς

“Less than a right [angle].” *ἐλάττων* describes the understood *γωνία*; *ὀρθῆς* is genitive after the comparison. The larger region contains the smaller angle.

### minor recto

Latin makes the same comparison with an ablative: “less than a right [angle].” *positus*, “placed,” agrees with masculine *angulus*. *in segmento maiore* places it in the larger region.

The complete printed sentence continues with the angle *of* a segment: Greek *τοῦ… τμήματος γωνία*, Latin *angulus segmenti*. That later statement concerns an angle made with the curved boundary. Here we measure the straight-chord angle *in* the segment; the two phrases are not interchangeable. Compare definitions III.6–8.

---

Read the instructions

# A compass and a carpenter’s square.

Doric · IV.3.9

### in medio autem quadrato circini centrum conlocetur et agatur linea rotundationis, quae quadrationis angulos tangat

“In the middle of the square let the compass’s centre be placed, and let a curved line be drawn which touches the square’s corners.”

The circle passes through all four corners. Its centre lies half a groove width above the mouth of the cut; the hollow uses a quarter of that circle. Vitruvius also permits twenty flat faces in this paragraph.

▼ Read the Latin construction

Ionic · III.5.14

### cacumenque normae circum rotundationem tangendo pervagari possit

“And the point of the square should be able to travel round the curve, keeping contact.”

The preceding words place the tool against the groove’s corners. With a 90° square, simultaneous contact with both edges is consistent with a semicircular hollow. This paragraph specifies twenty-four flutes.

▼ Read the Latin and its editorial supply

*in medio quadrato*: “in the middle of the square,” with the ablative after *in*. *circini* is genitive: “of the compass.” *centrum* is the subject of the passive command *conlocetur*, “let it be placed.”

*agatur*: “let [a line] be drawn.” *quae* refers back to *linea*; *angulos* is what that line touches. The quoted clause avoids the disputed bracketed word earlier in Krohn’s paragraph.

Inspect the complete Latin page

*normae* means “of the square,” a carpenter’s right-angle tool. *cacumen* is its point; *pervagari possit* means “should be able to move about.” *tangendo* expresses the action of keeping contact.

Krohn prints *anconibus 〈angulos〉 striarum dextra ac sinistra tangat*: “[it] should touch the groove’s corners on the right and left with its arms.” The brackets mark *angulos* as an editorial supplement; the apparatus credits Jocundus. Inspect the apparatus beneath the text.

Inspect the Latin and brackets

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A Greek word in its own context

# The fluting of a column.

## ἡ γὰρ τῶν λίθων σύνθεσις ἑτέρα τῆς τοῦ κίονος ῥαβδώσεως

“For the assembling of stones is different from the fluting of a column.”

Aristotle, *Nicomachean Ethics* X.4, 1174a23–24. He contrasts different activities within building. This is evidence for the word and its use, not an instruction for carving a profile.

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## τῆς (τοῦ κίονος) ῥαβδώσεως of the fluting of the column

---

### κίων → κίονος

“Column,” then “of a column.” The genitive *τοῦ κίονος* sits inside the larger noun phrase. Its article is masculine.

### ῥάβδωσις → ῥαβδώσεως

“Fluting,” then “of fluting.” The feminine article *τῆς* belongs to this outer noun, not to *κίονος*. Read the LSJ entry.

### ἑτέρα τῆς…

“Different from…” takes the genitive here. *σύνθεσις*, “assembling,” is the subject; *τῶν λίθων* tells us what is assembled: stones.

Find the Greek clause on the printed page

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Reopen these settings in Gonia. Source pages and fonts are embedded. Code, diagrams and new English: Anton Soloviev, 2026, MIT. Vitruvius, Aristotle and these historical printed pages: public domain. Fonts retain their licenses in the Gonia source repository.

Haec ratio in operibus diastylis erit constituta. si vero 7
systylon et monotriglyphon opus erit faciundum, frons
aedis, si tetrastylos erit, dividatur in partes XVIIII S, si
hexastylos erit, dividatur in partes XXVIIII S. ex his pars
una erit modulus, ad quem, uti supra scriptum est, di-
vidantur. ita supra singula epistylia et metopae et tri-
glyphi bini erunt conlocandi, in angularibus hoc amplius
dimidiatus et, quantum est spatium hemitriglyphi, id ac-
cedit. in mediano contra fastigium trium triglyphorum
et trium metoparum spatium distabit, quo latius medium
intercolumnium accedentibus ad aedem habeat laxamentum
et adversus simulacra deorum aspectus dignitatem.

Columnas autem striari XX striis oportet. quae si 9
planae erunt, angulos habeant XX designatos. sin autem
excavabuntur, sic est forma facienda, ita uti quam ma-
gnum est intervallum striae, tam magnis [striaturae] pa-
ribus lateribus quadratum describatur; in medio autem
quadrato circini centrum conlocetur et agatur linea ro-
tundationis, quae quadrationis angulos tangat, et quantum
erit curvaturae inter rotundationem et quadratam descrip-
tionem, tantum ad formam excaventur. ita dorica co-
lumna sui generis striaturae habebit perfectionem. de 10
adiectione eius, qua media adaugetur, uti in tertio volu-
mine de ionicis est perscripta, ita et in his transferatur.

Quoniam exterior species symmetriarum et corin-
thiorum et doricorum et ionicorum est perscripta, necesse
est etiam interiores cellarum pronaique distributiones ex-
plicare. distribuitur autem longitudo aedis, uti latitudo IV
sit longitudinis dimidiae partis, ipsaque cella parte quarta
longior sit, quam est latitudo, cum pariete, qui paries

1 diastylis *ed*: -liis *x* 3 tetrastylos *ed*: -lo *x* 3—4 XVIIII S . . . XXVIIII S *Barbarus*: XVIII . . . XXVIIII *x* 6 metopae *Joc*: metopha *x* 8 dimidiatus et quantum *Rode*: quantum dimidiatum *x* 9 (in mediano) habens — perpendiculum *repet. x* (*cf. supra*); *sed hic* moduli *om. HS* 10 quo *Lor*: quod *x* 11 habeat *om. G* (hebeat *in m. suppl.* $G^c$) laxiamentum *H* 16 *tit. secl. Kr* 18 centrum $SG^c$: centum *HG* 20 destructionē *a. c. G* 23 qua *Schn*: quae *x* 29 quarta parte *S*

Doric · Krohn (1912), p. 85. IV.3.9. Square, compass and twenty divisions. The complete page retains the critical apparatus. Library source.

pani autem, quod est in fastigio, altitudo sic est facienda,
uti frons coronae ab extremis cymatiis tota dimetiatur in 82
partes novem et ex eis una pars in medio cacumini tym-
pani constituatur, dum contra epistylia columnarumque
hypotrachelia ad perpendiculum respondeant. coronae- 5
que supra aequaliter imis praeter simas sunt conlocandae. (5)
insuper coronas simae, quas Graeci επαιετιδας dicunt, facien-
dae sunt altiores octava parte coronarum altitudinis. acro-
teria angularia tam alta, quantum est tympanum medium,
mediana altiora octava parte quam angularia. 10
13 Membra omnia, quae supra capitula columnarum sunt (10)
futura, id est epistylia, zophora, coronae, tympana, fastigia,
acroteria, inclinanda sunt in fronte suae cuiusque altitudinis
parte XII, ideo quod, cum steterimus contra frontes, ab oculo
lineae duae si extensae fuerint et una tetigerit imam operis 15
partem, altera summam, quae summam tetigerit, longior fiet. (15)
ita quo longior visus linea in superiorem partem procedit,
resupinatam facit eius speciem. cum autem, uti supra scrip-
tum est, in fronte inclinata fuerint, tunc in aspectu vide-
14 buntur esse ad perpendiculum et normam. columnarum 20
striae faciendae sunt XXIIII ita excavatae, uti norma in cavo (20)
striae cum fuerit coniecta, circumacta anconibus ⟨angulos⟩
striarum dextra ac sinistra tangat cacumenque normae cir-
cum rotundationem tangendo pervagari possit. crassitudines
striarum faciendae sunt, quantum adiectio in media columna 25
15 ex descriptione invenietur. in simis, quae supra coronam (25)
in lateribus sunt aedium, capita leonina sunt scalpenda,
disposita ⟨ita⟩, uti contra columnas singulas primum sint 83
designata, cetera aequali modo disposita, uti singula singulis
mediis tegulis respondeant. haec autem, quae erunt contra 30

---

2 ut *S* 3 cacumini *Kr*: -ne *x* 4 epistylia *Joc*: -lii *x* 7 epitidas *HG*, epytidas *S* (*corr. Böttticher*) 12 coronae *Joc*: corona & *x* 13 fronte *S*: frontis $HG^c$ 16 alterā *S et a. ras. H* 17 procedet *a. corr. G* 18 resopinatam *S et a. corr. H* 19 fuerint *Joc*: fuerit *x* 22 *add. Joc* (*post* sinistra) 23 cacumen-que *Ussing*: acumen quae *x* 24 rotundationem *rec*: -ne *x* crassitudinis *H* 28 *add. Ro* 30 autem *om. S*

**Ionic · Krohn (1912), p. 74.** III.5.14. XXIIII flutes and the norma. Krohn marks ⟨angulos⟩ as a supplement; his apparatus credits Jocundus. Library source.

IV. Τί δ' ἐστὶν ἢ ποῖόν τι, καταφανέστερον γένοιτ' ἂν ἀπ' 3
ἀρχῆς ἀναλαβοῦσιν. δοκεῖ γὰρ ἡ μὲν ὅρασις καθ' ὁντινοῦν
χρόνον τελεία εἶναι· οὐ γάρ ἐστιν ἐνδεὴς οὐδενὸς ὃ εἰς ὕστερον
γινόμενον τελειώσει αὐτῆς τὸ εἶδος· τοιούτῳ δ' ἔοικε καὶ
ἡ ἡδονή. ὅλον γάρ τι ἐστί, καὶ κατ' οὐδένα χρόνον λάβοι τις
ἂν ἡδονὴν ἧς ἐπὶ πλείω χρόνον γινομένης τελειωθήσεται τὸ
2 εἶδος. διόπερ οὐδὲ κίνησίς ἐστιν. ἐν χρόνῳ γὰρ πᾶσα κίνησις
καὶ τέλους τινός, οἷον ἡ οἰκοδομική, καὶ τελεία ὅταν ποιήσῃ
οὗ ἐφίεται. ἢ ἐν ἅπαντι δὴ τῷ χρόνῳ ἢ τούτῳ. ἐν δὲ τοῖς
μέρεσι καὶ τῷ χρόνῳ πᾶσαι ἀτελεῖς, καὶ ἕτεραι τῷ εἴδει τῆς
ὅλης καὶ ἀλλήλων. ἡ γὰρ τῶν λίθων σύνθεσις ἑτέρα τῆς
τοῦ κίονος ῥαβδώσεως, καὶ αὗται τῆς τοῦ ναοῦ ποιήσεως· καὶ
ἡ μὲν τοῦ ναοῦ τελεία (οὐδενὸς γὰρ ἐνδεὴς πρὸς τὸ προκείμε-
νον), ἡ δὲ τῆς κρηπῖδος καὶ τοῦ τριγλύφου ἀτελής· μέρους
γὰρ ἑκατέρα. τῷ εἴδει οὖν διαφέρουσι, καὶ οὐκ ἔστιν ἐν
ὁτῳοῦν χρόνῳ λαβεῖν κίνησιν τελείαν τῷ εἴδει, ἀλλ' εἴπερ, ἐν
3 τῷ ἅπαντι. ὁμοίως δὲ καὶ ἐπὶ βαδίσεως καὶ τῶν λοιπῶν. εἰ
γάρ ἐστιν ἡ φορὰ κίνησις πόθεν ποῖ, καὶ ταύτης διαφοραὶ
κατ' εἴδη, πτῆσις βάδισις ἅλσις καὶ τὰ τοιαῦτα. οὐ μόνον
δ' οὕτως, ἀλλὰ καὶ ἐν αὐτῇ τῇ βαδίσει· τὸ γὰρ πόθεν ποῖ
οὐ τὸ αὐτὸ ἐν τῷ σταδίῳ καὶ ἐν τῷ μέρει, καὶ ἐν ἑτέρῳ μέρει
καὶ ἐν ἑτέρῳ, οὐδὲ τὸ διεξιέναι τὴν γραμμὴν τήνδε κἀκείνην·
οὐ μόνον γὰρ γραμμὴν διαπορεύεται, ἀλλὰ καὶ ἐν τόπῳ 1174 b
οὖσαν, ἐν ἑτέρῳ δ' αὕτη ἐκείνης. δι' ἀκριβείας μὲν οὖν περὶ
κινήσεως ἐν ἄλλοις εἴρηται, ἔοικε δ'· οὐκ ἐν ἅπαντι χρόνῳ
τελεία εἶναι, ἀλλ' αἱ πολλαὶ ἀτελεῖς καὶ διαφέρουσαι τῷ

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16. γινόμενον $K^b$ : γενόμενον vulg. 17. ὅλον] τέλειον $K^b$ $M^b$ τι γάρ $K^b$ 20. ἡ om. $K^b$ $M^b$ καὶ add. $K^b$ 21. δὴ om. $K^b$ ἢ om. $L^b$ $M^b$ 22. καὶ τῷ χρόνῳ $K^b$ : τοῦ χρόνου vulg. (fort. τῷ χρόνῳ seludendum) 25. γὰρ om. $K^b$ 29. βαδίσεων $K^b$ $M^b$ 31. καὶ om. $K^b$ $M^b$ 34. ἐν add. $K^b$ 1174 b 1. γὰρ om. $K^b$

**Greek · Bywater, 1920 printing, p. 205.** Nicomachean Ethics X.4, 1174a23–24. The clause starts midway down the page, with ἡ γὰρ τῶν λίθων. This discusses different building activities, not flute geometry. Library source.

Ἔστω ἡ δοθεῖσα περιφέρεια ἡ ΑΔΒ· δεῖ δὴ τὴν ΑΔΒ περιφέρειαν δίχα τεμεῖν.

Ἐπεζεύχθω ἡ ΑΒ, καὶ τετμήσθω δίχα κατὰ τὸ Γ, καὶ ἀπὸ τοῦ Γ σημείου τῇ ΑΒ εὐθείᾳ πρὸς ὀρθὰς ἤχθω ἡ ΓΔ, καὶ ἐπεζεύχθωσαν αἱ ΑΔ, ΔΒ.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΑΓ τῇ ΓΒ, κοινὴ δὲ ἡ ΓΔ, δύο δὴ αἱ ΑΓ, ΓΔ δυσὶ ταῖς ΒΓ, ΓΔ ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ ΑΓΔ γωνίᾳ τῇ ὑπὸ ΒΓΔ ἴση· ὀρθὴ γὰρ ἑκατέρα· βάσις ἄρα ἡ ΑΔ βάσει τῇ ΔΒ ἴση ἐστίν. αἱ δὲ ἴσαι εὐθεῖαι ἴσας περιφερείας ἀφαιροῦσι τὴν μὲν μείζονα τῇ μείζονι τὴν δὲ ἐλάττονα τῇ ἐλάττονι· καί ἐστιν ἑκατέρα τῶν ΑΔ, ΔΒ περιφερειῶν ἐλάττων ἡμικυκλίου· ἴση ἄρα ἡ ΑΔ περιφέρεια τῇ ΔΒ περιφερείᾳ.

Ἡ ἄρα δοθεῖσα περιφέρεια δίχα τέτμηται κατὰ τὸ Δ σημεῖον· ὅπερ ἔδει ποιῆσαι.

## λα'.

Ἐν κύκλῳ ἡ μὲν ἐν τῷ ἡμικυκλίῳ γωνία ὀρθή ἐστιν, ἡ δὲ ἐν τῷ μείζονι τμήματι ἐλάττων ὀρθῆς, ἡ δὲ ἐν τῷ ἐλάττονι τμήματι μείζων ὀρθῆς· καὶ ἔτι ἡ μὲν τοῦ μείζονος τμήματος γωνία μείζων ἐστὶν ὀρθῆς, ἡ δὲ τοῦ ἐλάττονος τμήματος γωνία ἐλάττων ὀρθῆς.

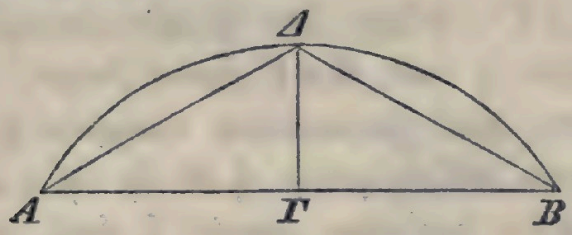
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XXXI. [Euclid.] opt. 47 (Studien p. 122). Alexander Aphrod. in metaph. p. 318. Simplicius in phys. fol. 14$^{u}$. Philop. in anal. II fol. 85$^{u}$. Boetius p. 388, 10.

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1. *ΑΔΒ*] litt. *ΔΒ* in ras. V; *ΑΒ* corr. ex *ΑΓ* P. 2. *ΑΒΔ* Bp; *ΑΒ* P. 3. *δίχα*] *ἡ ΑΒ δίχα* V. 5. *ΓΔ*] sic φ, e corr. m. 2 V. *καί*] om. φ. *ΔΒ*] *Β* corr. ex *Θ* m. 1 F. 8. *εἰσίν*] PBF; *εἰσί* Vp. 9. *καὶ βάσις* Bp, V m. 2. *ἄρα*] om. V. 10. *ἐστί* V. *δ' ἴσαι* V. 11. *ἀφαιροῦσιν* B; in

Sit datus arcus $A\Delta B$. oportet igitur arcum $A\Delta B$ in duas partes aequales secare.




ducatur $AB$ et in duas partes aequales secetur in $\Gamma$ [I, 10], et a puncto $\Gamma$ ad rectam $AB$ perpendicularis ducatur $\Gamma\Delta$, et ducantur $A\Delta$, $\Delta B$. et quoniam $A\Gamma = \Gamma B$, et communis est $\Gamma\Delta$, duae rectae $A\Gamma$, $\Gamma\Delta$ duabus $B\Gamma$, $\Gamma\Delta$ aequales sunt; et

$$\angle A\Gamma\Delta = B\Gamma\Delta;$$

nam uterque rectus est. itaque $A\Delta = \Delta B$ [I, 4]. uerum aequales rectae aequales arcus abscindunt maiorem maiori minorem autem minori [prop. XXVIII]. et uterque arcus $A\Delta$, $\Delta B$ minor est semicirculo. itaque arc. $A\Delta = \Delta B$.

Ergo datus arcus in duas partes aequales sectus est in puncto $\Delta$; quod oportebat fieri.

## XXXI.

In circulo angulus in semicirculo positus rectus est, qui autem in segmento maiore positus est, minor recto, qui autem in segmento minore positus est, maior recto, et praeterea angulus segmenti maioris maior est recto, minoris autem segmenti angulus minor recto.

ras. m. 1 P. 12. *ἐλάτονι* P. *ἑκατέρων φ.* *τῶν*] *τοῦ φ.* *ΔB*] om. F. 14. *ΔB*] in ras. V. *περιφερείᾳ*] om. V, *περιφέρειαν φ.* 15. *ἡ*] in ras. V. 16. *ποιῆσαι*] *δεῖξαι* P. 17. *λγ΄* F. 18. *ἐν*] post ras. 1 litt. V. 22. *γωνία*] m. 2 V. 23. *ὀρθῆς*] PF; *ἐστὶν ὀρθῆς* Bp; *ὀρθῆς ἐστιν* V.



**Euclid · Latin, Heiberg (1883), p. 241.** III.31 begins near the bottom. It distinguishes an angle IN a segment from the later angle OF a segment. Heiberg's Latin is an 1883 scholarly translation. The upper diagram belongs to the preceding III.30; our paired diagram is new. Library source.